O-C-110a
March 5, 1962

SUPERSEDING
Fed. Spec. O–C–110
August 21, 1952

FEDERAL SPECIFICATION

CALCIUM HYDROXIDE, TECHNICAL (HYDRATED LIME)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers one type and one grade of calcium hydroxide known commercially as "Hydrated Lime" (see 6.1).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS.

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

UU–S–48—Sacks, Shipping, Paper.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications, and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Ill.)

American Society For Testing and Materials Standards:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:12Z
Check the source to verify that this is the current version before use.

C 25–58—Methods of Chemical Analysis of Limestone, Quicklime, and Hydrated Lime.
C 50–57—Methods of Sampling, Inspection, Packing, and Marking of Lime and Limestone Products.
C 110–58—Methods of Physical Testing of Quicklime and Hydrated Lime.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa.)

3. REQUIREMENTS

3.1 When received the calcium hydroxide shall be in the form of fine white powder free from lumps and hard caking in the container.

3.2 **Calcium hydroxide content.** Calcium hydroxide content shall be no less than 90.0 percent by weight when tested in accordance with 4.6.2.

3.3 **Particle size.** The particle size shall be such that no less than 99.5 percent by weight shall pass through a No. 30 sieve and no less than 98.0 percent shall pass through a No. 200 sieve when tested in accordance with 4.6.3.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of section 5 have been satisfied.

4.3 **Lotting.** A lot shall consist of the calcium hydroxide produced by one manufacturer under essentially the same manufacturing conditions with no change in materials provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.4 **Sampling.**

4.4.1 *Nondestructive examination.* Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2 *For test.* Samples shall be taken from each delivery in accordance with the "Sampling" section of ASTM C 50–57.

4.5 **Examination.** Sample containers of calcium hydroxide shall be examined for level A and B preparation for delivery requirements in accordance with the classification of defects and with MIL–STD–105.

4.5.1 *Classification of defects.*

4.5.1.1 *Levels A and B packing* (section 5).

Categories	*Defects*
Critical:	None defined
Major:	*AQL 1.5 percent defective*
101	Shipping sack not as specified
102	Quantity packed not as specified or indicated
103	Shipping sack closure incorrect
104	Shipping sack damaged
105	Marking missing, incorrect, or incomplete

4.6 **Tests.** Distilled water and analytical reagent grade chemicals shall be used through-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:12Z
Check the source to verify that this is the current version before use.

out the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.6.1 *Visual.* The sample shall be visually examined for conformance with 3.1.

4.6.2 *Calcium hydroxide content.* The calcium hydroxide content shall be determined in accordance with ASTM C 25–58, rapid sugar method. This method determines the percentage of calcium oxide. To convert this figure to percent calcium hydroxide multiply by 1.321.

4.6.3 *Particle size.* The particle size shall be determined in accordance with ASTM C 110–58, residue of hydrated lime.

4.7 **Acceptance/rejection criteria.** If any sample fails to meet the test requirements of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packing shall be in accordance with Fed. Std. No. 102.

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 *Level A.* Calcium hydroxide shall be packed in 50-pound quantities in shipping packs conforming to construction number 2X of UU–S–48.

5.1.2 *Level B.* Calcium hydroxide shall be packed in 50-pound quantities in shipping packs conforming to construction number 3L/W of UU–S–48.

5.1.3 *Level C.* Calcium hydroxide shall be packed in accordance with the manufacturer's commercial practice to assure safe delivery at first destination for immediate use. Containers shall conform to Uniform Freight Classification Rules and other regulations as applicable to the mode of transportation.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.2.2 *Military agencies.* In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use. Calcium hydroxide covered by this specification is intended for use in the treatment of sewage and industrial and potable water, as well as for general use. Calcium hydroxide (hydrated lime) for structural use is not covered by this specification.

6.2 Ordering data. Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Level of packing required.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Military use. Items procured under this specification for military use are to be limited to the variety shown on the applicable military standard. Personnel of the military departments are requested to refer to these documents for guidance.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:
Army—C
Navy—Y
Air Force—MAA
MILITARY INTEREST:
Navy—MC

Copies of this specification may be purchased from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402

Source: https://assist.dla.mil – Downloaded: 2026-02-26T11:12Z
Check the source to verify that this is the current version before use.